

MAT 21C Discussion 4

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§I Warm-Up

(Discuss) Consider a power series

$$c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + \cdots.$$

Discuss what the phrase “the power series converges at $x = a$ ” means if a is a constant. If the series converges absolutely at $x = a$, then what can we say about the convergence of the series at $x = -a$? Does your argument work for negative a as well?

§II Power Series

Recall the very first discussion where I showed you the Taylor series for $\cos x$ which is

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}.$$

This is one of the tools to calculate $\cos x$ and get the computer to show you a number with finite precision on screen. The computer can not calculate the theoretical value of say $\cos 0.01$ because it would involve infinitely many calculations and the ability to show infinitely many digits.

There are two questions that need to be answered to be confident about this method.

- What is the interval of convergence of the power series?
- Does the power series converge to our target function on the interval of convergence?

Only if we have the answers to these two questions, we can be sure that the power series is a valid method to calculate the function.

There are only six possibilities for the interval of convergence of a power series

$$c_0 + c_1(x - a) + c_2(x - a)^2 + c_3(x - a)^3 + c_4(x - a)^4 + \cdots$$

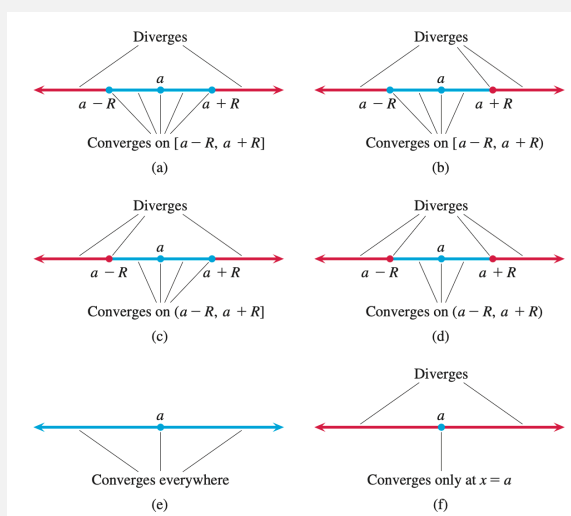


Figure 1: Six possibilities for the interval of convergence of a power series.

Moreover, the series always converges absolutely on the interior of the interval of convergence. This is helpful for determining the radius of convergence. In fact, I think the radius of convergence should be referred to as the radius of absolute convergence.

Finding the interval of convergence consists of three steps:

1. Apply the ratio or root test (or other methods that shows *absolute convergence*), so we are looking at the absolute convergence condition

$$\lim_{n \rightarrow \infty} \left| \frac{(n+1)\text{th term}}{n\text{th term}} \right| < 1 \quad \text{or} \quad \lim_{n \rightarrow \infty} \sqrt[n]{|n\text{th term}|} < 1.$$

2. Rewrite it as

$$|x - a| < R$$

for some R (possibly zero or infinity due to $\frac{1}{0} = \infty$ and $\frac{1}{\infty} = 0$), which is the radius of convergence.

3. If $R = 0$, then the interval of convergence is just $\{a\}$ (just a point which is technically still an interval). If $R = \infty$, then the interval of convergence is $(-\infty, \infty)$ (the whole real line). If R is a positive real number, then we have to check for convergence at the endpoints $a - R$ and $a + R$ to pin down the four possibilities: $(a - R, a + R)$, $[a - R, a + R)$, $(a - R, a + R]$, or $[a - R, a + R]$.

As an example, let us find the interval of convergence of the power series

$$1 - x^2 + x^4 - x^6 + x^8 - x^{10} + \dots = \sum_{n=0}^{\infty} (-1)^n x^{2n}.$$

One way is to notice that this is a geometric series with common ratio $-x^2$ so it converges absolutely if and only if $|-x^2| < 1$ which can be rewritten as $|x| < 1$. Applying the ratio test gives the same condition:

$$\lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} x^{2(n+1)}}{(-1)^n x^{2n}} \right| = \lim_{n \rightarrow \infty} |x^2| = |x|^2 < 1 \iff |x| < 1.$$

Therefore, the radius of convergence is $R = 1$. Now, we need to check for convergence at the endpoints $x = -1$ and $x = 1$. At $x = -1$, the series becomes

$$1 - 1 + 1 - 1 + 1 - 1 + \dots$$

which diverges, and at $x = 1$, the series also becomes

$$1 - 1 + 1 - 1 + 1 - 1 + \dots$$

which diverges. Therefore, the interval of convergence is $(-1, 1)$.

(Exercise) Find the interval of convergence of the following power series.

$$1. \quad 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

$$2. \quad x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \frac{x^9}{9} - \dots$$

Solution.

1. The radius of convergence is $R = \infty$ so the interval of convergence is $(-\infty, \infty)$.

2. The radius of convergence is $R = 1$ and the series converges at both endpoints by the alternating series test, so the interval of convergence is $[-1, 1]$.

Term by term differentiation and integration theorems in the textbook actually shows that radius of convergence does not change after differentiation or integration of power series. There is an interesting exercise in Professor's discussion worksheet that uses this fact to obtain a cool formula for π which is

$$\frac{\pi}{4} = \arctan 1 = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \dots$$

§III Taylor Series

We begin with the question whether we can rewrite a target function $f(x)$ as a power series

$$c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + \cdots.$$

We can obtain a “best guess” by matching all orders of derivatives at $x = 0$, and this best guess is called the Taylor series of $f(x)$ generated at $x = 0$. It is given by

$$p(x) = f(0) + f'(0)x + \frac{f''(0)}{2 \cdot 1}x^2 + \frac{f^{(3)}(0)}{3 \cdot 2 \cdot 1}x^3 + \frac{f^{(4)}(0)}{4 \cdot 3 \cdot 2 \cdot 1}x^4 + \cdots = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n.$$

We can convince ourselves that this power series matches all orders of derivatives at $x = 0$ by checking that

$$\begin{array}{lcl} p(x) = f(0) + f'(0)x + \frac{f''(0)}{2 \cdot 1}x^2 + \frac{f^{(3)}(0)}{3 \cdot 2 \cdot 1}x^3 + \frac{f^{(4)}(0)}{4 \cdot 3 \cdot 2 \cdot 1}x^4 + \dots & \implies & p(0) = f(0) \\ p'(x) = f'(0) + \frac{f''(0)}{1}x + \frac{f^{(3)}(0)}{2 \cdot 1}x^2 + \frac{f^{(4)}(0)}{3 \cdot 2 \cdot 1}x^3 + \dots & \implies & p'(0) = f'(0) \\ p''(x) = f''(0) + \frac{f^{(3)}(0)}{1}x + \frac{f^{(4)}(0)}{2 \cdot 1}x^2 + \dots & \implies & p''(0) = f''(0) \\ \vdots & & \vdots \end{array}$$

We can generalize this definition to Taylor series generated by any point $x = a$ instead of $x = 0$, and the formula is

$$f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f^{(3)}(a)}{3!}(x-a)^3 + \cdots = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

The Taylor series is only a best guess. It is only sometimes true that the Taylor series converges to the target function. We need to verify it using the Taylor's formula.

(Taylor's formula) The remainder term

$$R_n(x) = f(x) - \left(f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n \right)$$

after approximating $f(x)$ by the Taylor polynomial of degree n is bounded by

$$|R_n(x)| \leq \frac{M}{(n+1)!} |x|^{n+1}$$

if M is greater than or equal to $|f^{(n+1)}(t)|$ for all t between 0 and x . For example, if $|f^{(n+1)}(t)|$ is increasing, then we can take $M = |f^{(n+1)}(x)|$.

If we can show that

$$\lim_{n \rightarrow \infty} |R_n(x)| \leq \lim_{n \rightarrow \infty} \frac{M}{(n+1)!} |x|^{n+1} = 0,$$

then we can conclude that the Taylor series converges to $f(x)$ at x .

As an example, the Taylor series of $f(x) = e^x$ is (using the fact that $f^{(n)}(x) = e^x$ and thus $f^{(n)}(0) = 1$)

$$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots.$$

We know that the series converges for all x from the previous exercise, but does it converge to e^x ? This is what is going to be answered by the Taylor's formula.

I will show that the Taylor series converges to e^7 at $x = 7$. The remainder at $x = 7$ is

$$R_n(7) = e^7 - \left(1 + 7 + \frac{7^2}{2!} + \frac{7^3}{3!} + \cdots + \frac{7^n}{n!} \right)$$

which is bounded by

$$|R_n(7)| \leq \frac{M}{(n+1)!} 7^{n+1}$$

if we can find M such that $M \geq |f^{(n+1)}(t)|$ for all t between 0 and 7. Since $f^{(n+1)}(t) = e^t$ is increasing, we can take $M = e^7$. Therefore, we have

$$|R_n(7)| \leq \frac{e^7}{(n+1)!} 7^{n+1}.$$

By the ratio test, we have

$$\lim_{n \rightarrow \infty} \left| \frac{e^7 \cdot 7^{n+2}}{(n+2)!} \cdot \frac{(n+1)!}{e^7 \cdot 7^{n+1}} \right| = \lim_{n \rightarrow \infty} \frac{7}{n+2} = 0 < 1,$$

so then

$$\lim_{n \rightarrow \infty} |R_n(7)| \leq \lim_{n \rightarrow \infty} \frac{e^7}{(n+1)!} 7^{n+1} = 0,$$

and we conclude that the Taylor series converges to e^7 at $x = 7$.

(Exercise)

1. Show that the Taylor series of e^x converges to e^9 at $x = 9$. (Skip if you feel this is too easy)
2. Show that the Taylor series of e^x converges to e^x for all $x > 0$.
3. Show that the Taylor series of e^x converges to e^x for all $x < 0$.
4. Find the Taylor series of $\cos x$ and show that it converges to $\cos x$ for all x .